



**MAHARAJA INSTITUTE Of TECHNOLOGY
MYSORE**



**An Autonomous Institution
Affiliated to VTU, Belagavi**



Department of Electronics & Communication Engineering

CONTROL SYSTEMS [M23BEC404]

“ Industry Visit Report”

Assignment 1

Presented By:

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SECTION: 4C

Company Name: Tycoon Innovative Technology.

Tycoon Innovative Technology, is an Electronics Manufacturing Services (EMS) provider located in the **KIADB Industrial Area of Hootagalli, Mysore**. As an established contract manufacturer in this prominent industrial hub, the company specializes in comprehensive printed circuit board (PCB) assembly and complete turnkey manufacturing solutions. Their technical capabilities span a wide range of production needs, including Surface Mount Technology (SMT), Through-Hole, Mixed, and Ball Grid Array (BGA) assemblies. They are fully equipped to handle advanced processes like RoHS/Lead-free soldering and precise placement of fine-pitch components such as 0201 packages and QFNs. Beyond component assembly, Tycoon supports businesses throughout the entire production lifecycle by providing engineering design assistance, material sourcing, prototyping, and final box-build system integration.

Specialized Operational Capacities:

The facility is architected to execute complex, mixed-technology assemblies that seamlessly merge traditional hardware configurations with micro-scale modern components. Their production floor is structurally optimized to support the following operational workflows

- **High-Mix, Variable-Volume Production:** The infrastructure accommodates extreme high-mix, high-density component layouts. They easily balance low-mix runs of just a few intricate prototyping units up to highly repetitive mass assembly lines.
- **Diverse Circuit Board Mediums:** Fabrication and populating capabilities cover single-layer boards up to dense multi-layered arrangements. They manipulate rigid substrates, specialized flexible circuits, metal core PCBs optimized for heavy heat dissipation, and dense LED arrays.
- **System Integration & Testing:** Beyond the circuit board level, the company offers dedicated lines for complete product box-builds. This involves complex electromechanical sub-assembly, wire and cable harness integration, mechanical casing assembly, and final functional burn-in product testing under simulated workloads.

Strategic Sourcing & Engineering Services:

A core market differentiator for Tycoon Innovative Technology is its internal design engineering arm focused entirely on product lifecycle management and manufacturing cost reduction. They actively provide value and reliability engineering services aimed at optimizing new electronics for faster time-to-market. Specifically, their engineers specialize in component re-engineering to counter industry-wide component obsolescence, allowing businesses to upgrade legacy product designs without redesigning their entire system from scratch. This engineering support is backed by established sourcing channels that allow the company to manage turnkey material procurement

smoothly. By enforcing strict quality assurance guidelines and ship-to-stock logistics control, they ensure that finished electronic sub-assemblies can bypass traditional inbound inspection and drop directly into a client's just-in-time (JIT) warehouse inventory.

Date and Time of visit : 27th, May 2026.

Observed Process:

1. PCB Loder.
2. Laser Working.
3. Solder Past Apply on PCB board.
4. Component placing
5. Checking component was placed correctly.
6. Solder the components by heating (or) Heat Gun.
7. Inspect the PCB board.
8. Cleaning the PCB board.
9. Cutting the PCB boards.
10. Storing the boards.

Relevant photos:



Learning related to control systems:

Tycoon specializes in electronic manufacturing services (EMS), PCB assemblies, and hardware system integration. To build your expertise for roles like this, local training centers provide practical industrial automation programs.

- **System Architecture**: Describe how sensors monitor inputs (like temperature, pressure, or flow) and send signals to the main controllers.
- **Controller Implementation**: Focus on **Programmable Logic Controllers (PLCs)**. A standard control formulation might utilize proportional-integral-derivative (PID) control,
- **Supervisory Level**: Detail how **SCADA** or Distributed Control Systems (DCS) tie these controllers together to enable operators to supervise manufacturing processes.
- **Actuators and Final Control Elements**: Explain how the control system actuates physical changes (e.g., variable frequency drives (VFDs) altering motor speeds, or solenoid valves opening/closing).
- **Process Variables and Tuning**: Discuss how the control loop handles disturbances and how the PID parameters are tuned to achieve process stability.

Out comes of the visit:

The site visit to the Tycoon facility in Hebbal aimed to evaluate the health, automation efficiency, and security of the facility's Industrial Control System (ICS). The primary findings indicate robust foundational hardware, but note critical needs for software modernization, data latency mitigation, and cybersecurity reinforcement.

- **Hardware & Architecture:** The programmable logic controllers (PLCs) and distributed control systems (DCS) are operational and currently functioning at approximate 85% of capacity.
- **Data Latency:** A marginal delay < 150 ms was observed in the SCADA network during peak shift hours.
- **Safety Compliance:** Emergency shut-off systems and automated fail-safes are fully compliant with current industry safety standards.
- **Predictive Maintenance:** The control sensors lack predictive analytics, meaning minor thermal deviations are currently missed until they trigger manual alarms.

